ONE NUMBER PER CELL: A SEPARATING-FAMILY ENCODING OF SUPERIMPOSED DISCRETE CURVES

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Abstract. On a discrete grid several curves may cross and share cells, yet a tracer following one curve must recover, at every shared cell, where *its own* curve continues. We ask for a single static number per cell — no stored lists — from which each curve reads its next step with no collision. The answer is to read the cell number as a member of a *separating family*: assign curve i the prime p_i and store the product of p_i raised to curve i's local direction; each curve recovers its direction as the exponent of its own prime, and unique factorisation guarantees that distinct curves never interfere. We analyse its cost and give more compact separating families — residue (CRT) and positional codes — that share the same properties without factorisation. This is sharp. A local, state-blind tracer cannot resolve a tangency — two curves entering a cell alike but leaving differently — no matter how large the cell number: the local-indistinguishability theorem. The only resource that resolves it is a per-curve index, of order $\log m$ bits, carried in the tracer, not in the cell. The construction has a mutable dual: the rotor-router places the same disambiguating resource inside the cell and advances it on each visit. Separating family and rotor-router are the two escapes of the impossibility boundary — static cell with an indexed tracer, and mutable cell with a blind tracer.

1. Introduction

Draw several discrete curves on a grid and let them cross. Each cell that a curve passes through must say where that curve goes next; but a cell shared by two curves must serve both, and it may hold only *one* number. We do not allow the cell to store a list of (curve, direction) pairs — one static integer, nothing more — and we require that a tracer walking one curve, reading only the number at the cell it currently occupies, always recovers the continuation of *the curve it is following* and no other. At a shared cell the tracers of the several curves all read the same number and must each extract a different instruction from it, with no collision. This is the whole problem: *one number per cell, several superimposed curves, and each curve recovered locally at every shared cell.*[1]

Key words and phrases: digital geometry, chain codes, separating family, unique factorisation, rotor-router, discrete curves, one number per cell.

[1]An interactive demonstration — draw the curves and watch each code write one number per cell, including the closed and ρ-shaped curves of Remark 6.5 — runs in the browser at `https://l3ragio.github.io/one-number-per-cell/`.

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The answer. Read the cell's number as a member of a *separating family*. Give curve i its own prime p_i ($p_1 = 2, p_2 = 3, p_3 = 5, \dots$), let $d_i \in \{1, \dots, 8\}$ be curve i's outgoing direction in the Freeman chain code, and store at a shared cell z the single number

$$N(z) \;=\; \prod_i p_i^{\,d_i(z)}.$$

Curve i — knowing only that it is "the p_i curve" — recovers its own direction as the exponent of its own prime, the p_i-adic valuation $d_i(z) = v_{p_i}(N(z))$; every other curve is a different prime and is invisible to that reading. By unique factorisation the exponents never interfere, so the one number resolves the crossing with no collision (Definition 7.1 and Theorem 8.4). The prime code is one instance of a general algebraic pattern — a value space that factors, cell by cell, into one channel per curve — of which positional base-b codes and the Chinese Remainder Theorem are the additive and modular avatars (Definition 8.2 and Proposition 8.3).

The boundary. The construction is exactly as strong as it can be. If two curves enter a cell from the same direction, in the same tracer state, but must leave differently — a *tangency* — then no number stored at that cell, of any size, in any number system, tells the two apart: a single local deterministic tracer computes one output from the cell number and the entry, so it cannot return two different exits (the local-indistinguishability theorem, Theorem 10.14). The one resource that breaks the symmetry is the curve's own identity: a per-curve index, $\lceil \log_2 m \rceil$ bits for m curves, carried *in the tracer* rather than in the cell (Corollary 10.16). The separating family is precisely the statement that, once each tracer carries its index, the static cell number resolves every crossing.

Positioning. The static separating-family cell has a mutable dual: the *rotor-router* [HLM$^+$08], which stores one number per cell too but advances it on each visit, putting the disambiguating resource *inside the cell* rather than in the tracer's index — the two escapes of the same impossibility wall. Section 2 places the construction among chain codes, digital topology, per-cell position indices and navigation fields, self-organizing walkers, and separating encodings. This note was distilled from a line of work on threshold monitoring — the companion papers on observed implication [Bra26b], dual primitive modalities [Bra26c], and the intersection algebra of temporal property classes [Bra26a] — for which it supplies a geometric substrate; the genealogy is one-way and those papers do not depend on this one.

2. Related work

A single number written in each grid cell and read by a local tracer sits at the crossing of several established lines, in each of which either "one number per cell" or "a curve on a grid" already appears — but with a different meaning of the number. We locate the contribution against them.

Chain codes and digital curves. The direction code of Definition 6.1 is Freeman's chain code [Fre61, Fre74]: an integer per step naming a curve's local heading. Its first-difference form [Fre74] stores turning rather than absolute direction, and Bribiesca's vertex chain code [Bri99] counts contour vertices; all describe *one* curve as a *sequence*. Ours is spatial and superimposed: one cell, read in place, carries and separates the headings of m curves at once.

That a family of digital curves partitions the plane is the digital Jordan theorem of digital topology [Ros70, KR89], whose 4/8-connectivity pairing we invoke rather than reprove.

One number per cell, other senses. Two literatures already write a single number in every cell. Space-filling-curve indices — Morton/Z-order [Mor66] and the quadtree and Hilbert keys surveyed by Samet [Sam06] — encode *where* a cell is, a locality-preserving position label; ours encodes *what* passes through it. Numerical navigation functions and potential fields — the wavefront field [BLL92], artificial potentials [Kha86], and minimum-free navigation functions [RK92] — store a scalar distance-to-goal per cell that a *single* agent descends; ours stores a per-curve *direction* that m curves read independently, its correctness a separating-family property, not a Morse-theoretic one.

Superposition versus collision. Multi-agent path finding places many paths on one grid but treats a shared cell as a *conflict* to forbid and search away [SSFS15, SSF$^+$19]. We invert that axiom: a shared cell is a *superposition* to admit and algebraically separate, so no coordination search is needed to keep the curves apart.

Separating encodings. The value space of Definition 8.2 is a direct-sum code. Its multiplicative prime form is Gödel's sequence numbering [Göd31] and its residue form is Garner's residue number system [Gar59]; these, with mixed-radix codes and perfect hashing [FKS84], reappear in Section 9, where the prime form is shown dominated. Pairing functions [Ros03] likewise inject a tuple into one number but lack the cheap per-coordinate projections that curve separation needs.

The mutable-cell dual. The closest kin keep the cell *mutable* and the tracer index-blind: rotor-routing and Eulerian walkers [HLM$^+$08, PDDK96], the abelian sandpile [BTW87, Dha90], stigmergic ant-robots [WLB99], and Langton's ant [Lan86]. They put the disambiguating state *in the cell* and advance it on each visit; we keep the cell static and put the index *in the tracer*. Section 11 develops this duality with the rotor-router as the mutable escape of the same impossibility wall.

3. The grid and its values

There are two ways to put geometry on a discrete grid: sample a real magnitude against a family of cuts, so that a cell is an interval and geometry is the pattern of cut-crossings, or write the geometry directly into the cells. We take the second route — we discretise *both* axes at once and stop treating the cell content as a sampled number: the cell content becomes a *symbol* that names a local geometric fact.

Definition 3.1 (Grid and configuration)**.** A *grid* is a rectangular array $G = \{0, \dots, n-1\} \times \{0, \dots, m-1\}$ of *cells* (or the full lattice $\mathbb{Z}^2$). A finite *value alphabet* A carries a distinguished symbol $0 \in A$, read as *empty*. A *configuration* is a map

$$f \colon G \longrightarrow A,$$

and the *support* of f is the set of nonzero cells $\operatorname{supp}(f) = \{c \in G : f(c) \neq 0\}$.

Definition 3.2 (Neighbourhoods)**.** For a cell $c = (x, y)$, the 4-*neighbourhood* (von Neumann) is

$$N_4(c) = \{(x \pm 1, y),\ (x, y \pm 1)\},$$

the four edge-adjacent cells; the 8-*neighbourhood* (Moore) is

$$N_8(c) = N_4(c) \cup \{(x \pm 1, y \pm 1)\},$$

the eight cells sharing an edge or a corner. These are the two standard lattice adjacencies of cellular-automaton and image models *[von Neumann,* Theory of Self-Reproducing Automata*, the five-cell neighbourhood; Moore, the nine-cell neighbourhood].*

Remark 3.3 (Two discretisations of the same plane)**.** One discretisation reads a continuous value axis $\mathbb{R}$ against a finite family of cuts: the cell is an interval, no cell holds geometry, and geometry is the pattern of crossings of the cut lines. We take the dual. Here the value axis is itself the finite alphabet A, and the geometry is *written into* the cells: a cell's symbol is the geometric datum, not a quantised magnitude. This is the computer-vision / cellular-automaton discretisation, dual to the cut-based one. We call the resulting object a *self-descriptive grid*: the data *are* the picture, and (for curves, Section 6) the data even say how to walk the picture.

4. The point

The zeroth stratum is a single occupied cell with nothing around it.

Definition 4.1 (Point)**.** A *point* of a configuration f is a cell c with $f(c) \neq 0$ whose entire Moore neighbourhood is empty:

$$f(c) \neq 0 \qquad \text{and} \qquad f(c') = 0 \ \text{ for every } c' \in N_8(c).$$

It is an *isolated* nonzero cell in the Moore topology.

Proposition 4.2 (Point = 0-dimensional digital object)**.** A point has no digital adjacency to any other nonzero cell; it is a connected component of $\operatorname{supp}(f)$ of size 1 under 8-connectivity.

Proof. By Definition 4.1 every $c' \in N_8(c)$ is empty, so c is 8-adjacent to no cell of $\operatorname{supp}(f)$. Hence $\{c\}$ is closed under the 8-adjacency relation restricted to the support, i.e. a component of size 1. □

Remark 4.3 (Connectivity is digital adjacency)**.** Connectivity here is *digital* adjacency in the sense of digital topology *[Rosenfeld,* Digital topology*, 1979; Kong–Rosenfeld,* Digital topology: introduction and survey*, CVGIP 48, 1989 — metadata to be verified at deposit]*, not the metric connectedness of the Euclidean plane. The choice to include the four diagonal cells in the isolation condition is precisely the choice of 8-connectivity for the *foreground* (the nonzero cells); its complement, the background, is then read with the paired 4-connectivity. The pairing is not cosmetic — it is the exact device that makes the digital Jordan theorem hold.

Figure 1 shows, in one grid, an isolated point, a short segment, and a full line, with the empty cells written out as 0.

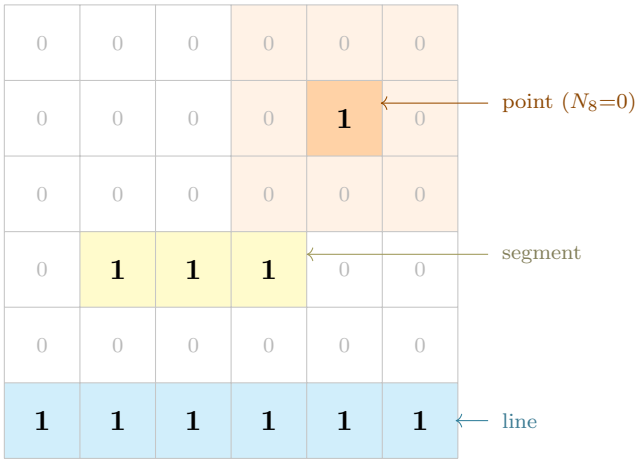


Figure 1: The three lowest strata on one 6×6 self-descriptive grid; empty cells carry the symbol 0. **Point** (orange, Definition 4.1): an isolated nonzero cell whose entire Moore neighbourhood N_8 (light orange) is empty — a 0-dimensional digital object. **Segment** (yellow, Definition 5.1): a maximal collinear run of 8-adjacent nonzero cells, here horizontal of length 3. **Line** (cyan, Definition 5.2): the segment that spans the whole grid extent in its direction, its endpoints on the boundary. All nonzero cells carry the generic occupancy value 1; the direction code of Section 6 is not yet in play.

5. The segment and the line

The point extends by adding a collinear neighbour. There is exactly one extension operator, and iterating it up to the grid boundary produces the line.

Definition 5.1 (Segment)**.** A *segment* is a maximal set $S \subseteq \mathrm{supp}(f)$ of nonzero cells that are pairwise 8-adjacent along a single grid direction and *collinear* in that direction — horizontal, vertical, or one of the two diagonals. Formally, there is a fixed step vector $u \in \{(1,0),(0,1),(1,1),(1,-1)\}$ and cells $c_0, c_1, \ldots, c_k$ with $c_{i+1} = c_i + u$ and $f(c_i) \neq 0$ for all i, maximal under this description. A segment has a *support direction* u and two *endpoints* — the cells with exactly one nonzero neighbour along u.

Definition 5.2 (Line)**.** A *line* is a segment that spans the whole grid extent in its support direction: its two endpoints lie on the grid boundary. It is the grid-filling segment for its direction.

Proposition 5.3 (The extension operator)**.** Every segment is obtained from a single point by finitely many applications of the operator *"adjoin a collinear 8-adjacent nonzero cell at an endpoint,"* and a line is a segment for which the operator can no longer be applied without leaving the grid.

Proof. Order the cells of a segment along its step vector u as $c_0, \ldots, c_k$. Starting from $\{c_0\}$ (a point in the degenerate $k = 0$ case) and adjoining $c_1, \ldots, c_k$ in turn realises S by k applications of the operator, each preserving collinearity and 8-adjacency by construction. Maximality of S (Definition 5.1) is exactly the statement that no further collinear nonzero cell is adjacent at either endpoint; when both endpoints are on the boundary the grid itself blocks further extension, which is Definition 5.2. □

Remark 5.4 (The dimension lift, purely combinatorial)**.** The strata rise by dimension: point (0-dimensional, isolated) → segment (1-dimensional, bounded) → line (1-dimensional, grid-spanning). The mechanism is entirely combinatorial: there is no operator on a value axis, no threshold, no intermediate-value law. The geometry lives in the *adjacency graph* of the nonzero cells, and the extension operator — adjoin a collinear neighbour — is a purely graph-theoretic move, walking the Moore adjacency.

6. The directed curve

Segment and line still store only *presence*: a nonzero cell says "the object passes through me" but not *where it goes*. The curve is where the self-descriptive idea does its real work: we let the cell value name the direction to the successor, so that reading a cell tells you where to step next.

Definition 6.1 (Freeman direction code)**.** Number the eight Moore neighbours $1, \dots, 8$ counterclockwise from east:

$$1{=}\mathrm{E}, \quad 2{=}\mathrm{NE}, \quad 3{=}\mathrm{N}, \quad 4{=}\mathrm{NW}, \quad 5{=}\mathrm{W}, \quad 6{=}\mathrm{SW}, \quad 7{=}\mathrm{S}, \quad 8{=}\mathrm{SE},$$

with the step vectors $\mathrm{dir}(1) = (1,0)$, $\mathrm{dir}(2) = (1,1)$, $\mathrm{dir}(3) = (0,1)$, $\mathrm{dir}(4) = (-1,1)$, $\mathrm{dir}(5) = (-1,0)$, $\mathrm{dir}(6) = (-1,-1)$, $\mathrm{dir}(7) = (0,-1)$, $\mathrm{dir}(8) = (1,-1)$. The symbol 0 names *no successor*. This is the Freeman chain code [Fre61], with the sole cosmetic change that Freeman's codes $0, \dots, 7$ are shifted to $1, \dots, 8$ so that 0 is reserved for the empty cell.

Definition 6.2 (Self-descriptive curve)**.** A *curve* is a configuration in which each cell's value encodes the direction to the next cell of the curve: $f(c) = d \in \{1, \dots, 8\}$ means *"the curve continues from c to the neighbour $c + \mathrm{dir}(d)$,"* and $f(c) = 0$ means c is not on the curve, or is its terminus. A curve is therefore a *self-describing path*: the value read at a cell is exactly the instruction for the next step,

$$c \;\longmapsto\; c + \mathrm{dir}\big(f(c)\big) \qquad \text{until } f(\,\cdot\,) = 0.$$

Proposition 6.3 (Curves carry intrinsic orientation)**.** A curve is *directed*: it is traversed in a definite orientation, unlike the undirected segment and line of Section 5. Reversing a curve requires recoding every cell with the reverse direction.

Proof. The code of Definition 6.1 names a *successor* direction, so the map $c \mapsto c + \mathrm{dir}(f(c))$ is a well-defined partial successor function on the cells; its transitive closure is a directed path. A segment or line (Definitions 5.1 and 5.2) stores only presence and induces the *symmetric* 8-adjacency relation, which carries no orientation. To reverse a curve one must replace, at each cell c with successor $c' = c + \mathrm{dir}(d)$, the value at c' by the code of the opposite step dir^{-1} pointing back to c; since $\mathrm{dir}(d)$ and its negation are distinct codes (the antipodal pair $d \mapsto d + 4 \bmod 8$ on the compass), this is a genuine recoding, not a relabelling. □

Remark 6.4 (The curve generalises the line)**.** A straight segment in direction d is exactly the constant-value curve $f \equiv d$: every cell says "keep going the same way." Lines and segments are thus the *constant* curves; a general curve *bends* as its value varies from cell to cell. This is the precise sense in which the curve generalises the line here — same substrate, same successor mechanism, only the direction code is now spatially varying rather than constant. In this discrete, self-descriptive setting a line is exactly a curve that never turns, with the motion stored *in the data* rather than read off a value axis.

Figure 2 draws a bent curve as cells carrying their direction digits, with an arrow in each cell pointing to its successor, and a Freeman compass inset.

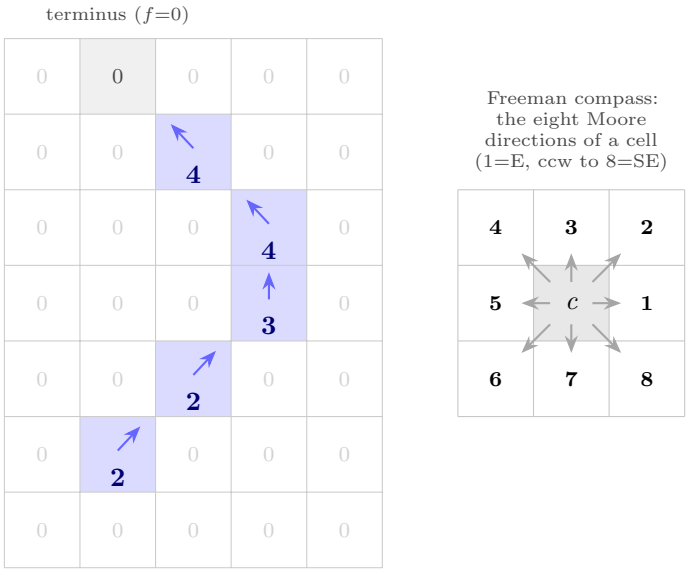


Figure 2: A self-descriptive directed curve (Definition 6.2) with cell values $2, 2, 3, 4, 4$ read along its path (blue), on a 5×7 grid; empty cells carry 0. Each digit is a Freeman direction (Definition 6.1); the blue arrow in each cell points to the successor cell $c + \mathrm{dir}(f(c))$: two steps NE (2), one step N (3), two steps NW (4), reaching the terminus cell where $f = 0$. The straight sub-runs are the constant-value stretches (Remark 6.4); the curve bends exactly where the digit changes. **Inset:** the Freeman compass, numbering the eight Moore neighbours $1, \ldots, 8$ counterclockwise from east.

Remark 6.5 (Closed curves and ρ-shapes)**.** The code is single-valued — $c \mapsto c + \mathrm{dir}(f(c))$ is a partial function — so a curve is any orbit of it, and the datum needs no change to carry more than an open path. Besides the path that halts at a terminus ($f = 0$), one direction per cell represents the *closed curve* (a cycle with no terminus: the eight border cells of a 3×3 block carry $1, 1, 3, 3, 5, 5, 7, 7$ counterclockwise and cycle forever) and the *ρ-shape* (a tail running into such a cycle, the trajectory meeting a cell it has visited). Each occupied cell keeps exactly one successor and nothing is stored twice, so the prime and residue codes apply verbatim, with no collision. The sole configuration beyond reach is a cell with *two* successors — a self-intersection, the curve crossing its own line — excluded by Theorem 10.14 and the object of the branch-jet extension (Section 12).

7. Superimposed curves and the shared cell

A single cell can lie on several curves, but one direction digit encodes only one successor. We need cell values in which several curves' directions coexist without interfering, and from which each curve's local direction can be recovered. Positional base-b codes give the additive instance — place each curve's direction in its own base-b digit column, carry-free

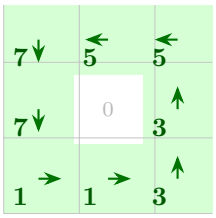


Figure 3: A closed curve, no new machinery (Remark 6.5): each of the eight border cells of a 3×3 block carries one Freeman code (Definition 6.1) and the successor map $c \mapsto c + \mathrm{dir}(f(c))$ runs round the loop with *no* terminus — counterclockwise from the lower-left, $1, 1, 3, 3, 5, 5, 7, 7$; the centre cell is empty (0).

when b exceeds the number of directions. We feature instead the multiplicative *prime* form: unbounded in the number of curves, it is the cleanest route to the separating-family algebra of Section 8.

Definition 7.1 (Gödel/prime code)**.** Assign curve i the i-th prime p_i ($p_1 = 2, p_2 = 3, p_3 = 5, \dots$). A cell z shared by curves $i_1, \dots, i_r$ receives the aggregate value

$$N(z) \;=\; \prod_{j=1}^{r} p_{i_j}^{\,d_{i_j}},$$

with $d_{i_j} \in \{1, \dots, 8\}$; an absent curve contributes exponent 0, and an empty or terminal cell — on which no curve continues — carries 0, the same no-successor symbol as the single-curve code f (Definition 6.1). Thus $N(z)$ is the whole-cell number (as in the introduction and Definition 8.2), distinct from the single-curve direction code f.

Proposition 7.2 (Prime decoding)**.** The exponent of p_i in $N(z)$ is exactly curve i's local direction through z (exponent 0 meaning "curve i absent"). The primes are non-interfering classes: unique factorisation guarantees no cross-talk between curves.

Proof. By the fundamental theorem of arithmetic the factorisation of $\prod_j p_{i_j}^{\,d_{i_j}}$ into primes is unique, so the exponent of each p_i is well defined and recovers d_i. This is precisely Gödel numbering [Göd31] applied to curve identity: the "sequence" being encoded is the vector of per-curve directions at the cell. □

8. A Separating-Family Algebra for Superimposed Curves

The base-9 and prime codes of Section 7 are two instances of one algebraic principle, and it pays to state the principle directly. We want a single algebraic system in which several curves may cross and yet every cell carries all the information needed to continue each of them. The move is to read a cell's number as a member of a *family* of functions that, place by place, encode distinguishably the datum of each curve, so that one projection per curve reads off that curve's local instruction and no other's.

Definition 8.1 (Local data)**.** At a cell z, a curve i passing through z has *local data* $d_i(z) \in D_i$, a member of a finite set D_i — its outgoing direction, or its (in, out) pair when entry matters.

Definition 8.2 (Separating family / uniquely-decodable superposition)**.** A *separating family* for curves $1, \dots, m$ is a value space V together with a *superposition* $\sigma\colon \prod_{i=1}^{m} D_i \to V$ and *projections* $\pi_i\colon V \to D_i$ such that

$$\pi_i\big(\sigma(d_1, \dots, d_m)\big) = d_i \qquad \text{for every } i \text{ and every } (d_1, \dots, d_m).$$

The cell stores $N(z) = \sigma\big(d_1(z), \dots, d_m(z)\big)$, and curve i recovers its own datum by $d_i(z) = \pi_i(N(z))$. The family *separates* the curves: each projection isolates one curve's data regardless of the others.

Proposition 8.3 (The classical codes are separating families)**.** Each of the following is a separating family; the prime code (Definition 7.1) of Section 7 is the multiplicative instance (b), and positional base-b codes are the additive instance (a).

(a) *Positional, base $b > \max_i |D_i|$:* $\sigma = \sum_i d_i\, b^{i-1}$, and π_i is the i-th base-b digit.

(b) *Gödel / prime:* $\sigma = \prod_i p_i^{d_i}$, and π_i is the p_i-adic valuation.

(c) *CRT:* pairwise-coprime moduli $m_i > |D_i|$, $V = \mathbb{Z}/(\prod_i m_i)$, σ the CRT combination ($N \equiv d_i \bmod m_i$), and π_i reduction mod m_i. The Chinese Remainder Theorem *is* the separation statement *[classical; the Chinese Remainder Theorem]*.

(d) *Orthogonal signal:* an orthogonal system $\{\varphi_i\}$, $\sigma = \sum_i d_i \varphi_i$, and $\pi_i = \langle \cdot, \varphi_i \rangle / \langle \varphi_i, \varphi_i \rangle$.

Abstractly all four are one *direct-sum / product* decomposition $V \cong \prod_{i=1}^{m} D_i$, one summand per curve.

Proof. (a) is base-b uniqueness; (b) is unique factorisation (Proposition 7.2). (c): the residue systems to pairwise-coprime moduli are in bijection with $\mathbb{Z}/(\prod_i m_i)$ by the Chinese Remainder Theorem, and $\pi_i = (\cdot \bmod m_i)$ inverts the i-th coordinate, the hypothesis $m_i > |D_i|$ embedding D_i in $\mathbb{Z}/m_i$. (d): orthogonality gives $\langle \sigma, \varphi_i \rangle = d_i \langle \varphi_i, \varphi_i \rangle$, so π_i recovers d_i. In each case σ is injective with left inverse $(\pi_1, \dots, \pi_m)$, i.e. a decomposition $V \cong \prod_i D_i$. □

Theorem 8.4 (Static resolution by separating families)**.** *Let m curves pairwise share cells. They are traceable by a* single static number per cell *— no cell mutation, no rotor — if and only if their local-data map $z \mapsto (d_1(z), \dots, d_m(z))$ factors through a separating family $V \cong \prod_i D_i$; and then the minimum $|V|$ is $\prod_i |D_i|$. Each tracer carries only its family index i — which curve it is — and reads $d_i(z) = \pi_i(N(z))$.*

Proof. ($\Leftarrow$) A separating family supplies σ and the π_i; store $N(z) = \sigma(d_1(z), \dots, d_m(z))$ and let the tracer of curve i output $\pi_i(N(z))$. Each cell holds one static number, and by Definition 8.2 every curve reads its own datum with no interference. ($\Rightarrow$) A static resolver for the family is, at each cell, a rule that from $N(z)$ and the index i returns $d_i(z)$; fixing i this is a map $\pi_i\colon V \to D_i$ with $\pi_i(N(z)) = d_i(z)$, and since $N = \sigma(\vec{d})$ must determine each coordinate, the π_i jointly invert σ — a separating family. The image must distinguish all $\prod_i |D_i|$ combinations of local data, so $|V| \geq \prod_i |D_i|$, attained by the positional code (a). □

Corollary 8.5 (Consistency with local indistinguishability)**.** *The family index i is exactly the* context *that the impossibility corollary (Corollary 10.16) requires. A tangency — same entry, and index-blind — is unresolvable by Theorem 10.14 precisely because it lacks the index; the separating family supplies the resolution once the index is known. Thus $\lceil \log_2 m \rceil$ bits of index in the tracer, plus the superposition in the cell, resolve every crossing and plateau statically.*

Remark 8.6 (The plateau, resolved statically)**.** On a shared plateau each cell holds $\sigma(d_1, \ldots, d_m)$; curve i follows π_i throughout the run and, at the branch cell, reads its own divergent direction d_i — the plateau is resolved by the direct-sum structure, with *no* mutable rotor. This is the exact *dual* of the rotor-router escape (Definition 11.5 and Proposition 11.6): the two escapes of the local-indistinguishability theorem are the *mutable cell with an index-blind tracer* — the rotor-router — and the *static cell with an indexed tracer* — the separating-family algebra. The rotor moves the resource into the cell over time; the separating family keeps the cell static and moves the resource into the tracer's index. The separating-family algebra is the purely algebraic realisation of the "one number per cell" intuition, resolving crossings and plateaus with no cell ever mutating.

Figure 4 shows one crossing cell whose single number is decoded by three separating families into the same two per-curve directions.

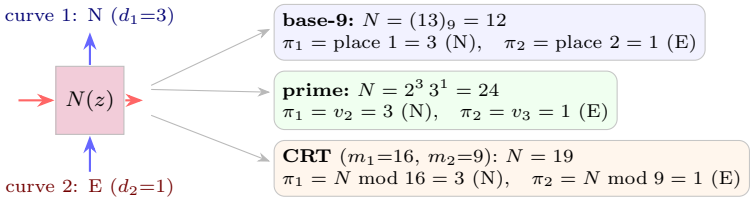


Figure 4: One number, three separating families, the same demultiplexed curves. The crossing cell holds a single number $N(z)$; three separating families (Definition 8.2) each decode *their* number into the *same* two per-curve directions — curve 1 north ($d_1 = 3$) and curve 2 east ($d_2 = 1$) — by their projections π_i: base-9 digit places, prime exponents (v_p the p-adic valuation, Definition 7.1), or residues modulo pairwise-coprime moduli (CRT). Each is the direct-sum decomposition $V \cong \prod_i D_i$ of Theorem 8.4 read in a different basis.

The point of the separating family is easy to miss in the abstract, so it is worth watching it fail and succeed on a single crossing. The following example contrasts two prime encodings of the *same* configuration; the contrast is the sharpest way to see *why* the family index of Theorem 8.4 must be attached to the *curve* and never to the *direction*.

Example 8.7 (Two prime encodings: directions vs. curves)**.** At a cell z two curves cross. Curve 1 leaves north (local datum $d_1 = 3$ in the Freeman code of Definition 6.1) and curve 2 leaves east ($d_2 = 1$). We want to store, in one number $N(z)$, exactly enough to tell each tracer where *its own* curve continues. There are two temptations, and only one works.

Encoding A — a prime per *direction* (the natural first attempt, and why it fails). Give each of the eight Moore neighbours its own prime, $q_{\mathrm{N}}, q_{\mathrm{E}}, \ldots$, one for each direction, and let the cell hold the *product* of the primes of the directions actually taken:

$$N_A(z) \;=\; q_{d_1}\, q_{d_2} \;=\; q_{\mathrm{N}}\, q_{\mathrm{E}}.$$

Factoring $N_A(z)$ recovers the multiset of exit directions present at z — here $\{\mathrm{N}, \mathrm{E}\}$ — and nothing more. It does *not* recover the assignment curve $\mapsto$ direction: the factor q_{N} is labelled by the *direction* north, not by any curve, so a tracer following curve 1 has no way to decide whether it should take north or east. The product is symmetric in the curves; it discards their identity. Worse, when both curves leave the same way — a plateau step — one gets $N_A = q_d^2$, whose multiplicity counts *how many* curves go that way while naming *none*.

Encoding A answers "which directions leave z" but never "which curve leaves which way": it is exactly the hypothesis-failure of the local-indistinguishability theorem (Theorem 10.14), the per-curve index simply absent.

Encoding B — a prime per *curve*, the direction in the *exponent* (the separating family). Move the prime off the direction and onto the curve: give curve i the prime p_i, and store

$$N_B(z) \;=\; \prod_i p_i^{d_i}, \qquad d_i \in \{1,\dots,8\} \text{ (with 0 for "curve absent")},$$

where the *exponent* d_i is curve i's Freeman direction. Here

$$N_B(z) \;=\; p_1^{d_1}\, p_2^{d_2} \;=\; p_1^3\, p_2^1 \qquad (\text{north} = 3,\ \text{east} = 1).$$

Curve i — knowing only that it is "the p_i curve" — recovers its own direction as the p_i-adic valuation $d_i = v_{p_i}\big(N_B(z)\big)$; every other curve is a different prime, invisible to that projection. The Freeman directions are the *exponents*; the primes label the *curves*. This is precisely the Gödel/prime instance of the separating family [Göd31] (Proposition 8.3(b)), and by Theorem 8.4 it resolves every crossing. It resolves a plateau for free as well: two curves leaving the same direction d give $p_i^d\, p_j^d$, each read off by its own prime with no collision — the static resolution of Remark 8.6, where Encoding A's q_d^2 had conflated them.

Moral. The Freeman directions must sit on the exponents and the primes must label the curves; putting the primes on the directions recovers the set of exits but provably not the curve-to-exit assignment, because the family index the theorem requires is the curve, not the direction.

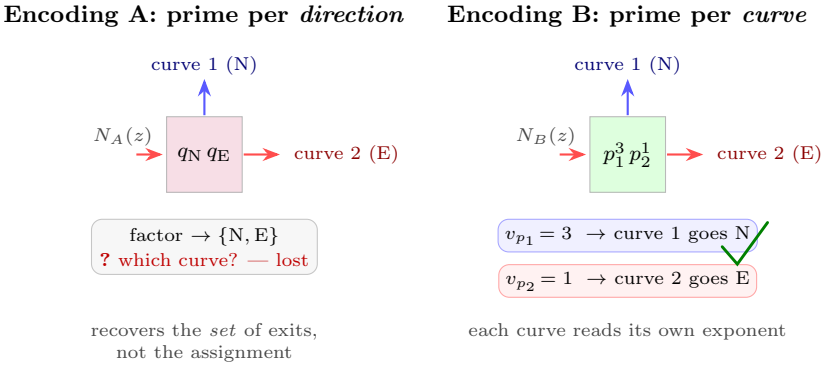


Figure 5: The same crossing cell z under two prime encodings (Example 8.7). **Left, Encoding A** puts a prime on each *direction* and stores the product $N_A = q_N q_E$; factoring returns only the *set* of exits $\{N, E\}$, so the question "which curve?" is lost — the hypothesis-failure of Theorem 10.14. **Right, Encoding B** puts a prime on each *curve* and the Freeman direction in the *exponent*, $N_B = p_1^3 p_2^1$; each curve reads its own direction off its own prime by the p_i-adic valuation ($v_{p_1} = 3 \Rightarrow$ N, $v_{p_2} = 1 \Rightarrow$ E), the separating family of Proposition 8.3(b). The directions belong on the exponents; the primes label the curves.

9. Efficiency: the Cost of the Prime Encoding and More Compact Separating Families

The prime code of Definition 7.1 is the cleanest *illustration* of a separating family — its separation is visible to the eye, since unique factorisation reads each curve off its own prime — but it is not the cheapest way to realise one. Two costs are usually held against it, and only one of them is real. This section separates the two, states each cost exactly, and then gives three drop-in alternatives that keep every property the resolution theorem (Theorem 8.4) needs while spending far less. Throughout, D denotes the size of the direction alphabet (for a two-dimensional Moore grid $D = 8$, and $d_i \in \{1, \dots, D\}$ is curve i's local direction) and m the number of curves through a cell.

9.1. **Which properties are essential, and which are bought back.**

The separating-family requirement is not monolithic. It decomposes into a handful of atomic properties, and efficiency is bought precisely by relaxing the ones that are inessential to a given deployment.

Definition 9.1 (Atomic properties of a shared-cell encoding)**.** An encoding $N(\cdot)$ of m superimposed curves is analysed along the seven atomic properties of Table 1. Properties (A1), (A2), (A7) are *essential* — an encoding lacking any one of them fails to be a separating family in the sense of Definition 8.2 — while (A3) and (A4) are *relaxable*: giving them up is legitimate whenever the deployment permits, and it is exactly where the cost is recovered.

Property	Meaning	Status
(A1) Separation	the superposition σ is injective, with per-curve projections π_i (unique decodability).	ESSENTIAL
(A2) Locality	$N(z)$ alone — plus the reader's own curve index — determines the step; no neighbour reads.	ESSENTIAL
(A3) Unbounded m	any number of curves may share a cell.	RELAXABLE
(A4) Per-curve random access	recover d_i without decoding the other $m-1$ curves.	RELAXABLE
(A5) Static cell	the datum does not mutate.	kept
(A6) Dimension-agnostic	works for any direction alphabet $D = 3^\delta - 1$ on a δ-dimensional grid.	desirable
(A7) Relabelling-symmetry	$N(z)$ is a superposition symmetric under curve relabelling given the index map (a product / direct sum).	ESSENTIAL

Table 1: The atomic properties of a shared-cell separating encoding (Definition 9.1). (A1), (A2), (A7) are essential to the separating-family property of Definition 8.2; (A3) and (A4) are relaxable and are exactly where the alternatives of Section 9.3 recover their cost. (A5) is kept here (the mutable rotor is the alternative of the next section); all candidates below satisfy (A6).

Every candidate below realises the *same* direct sum $V \cong \prod_i D_i$ — the separating family of Definition 8.2 — and hence resolves every crossing by Theorem 8.4. They differ only in the *coordinate system* placed on that product, and therefore only in cost.

9.2. **The cost of the prime-power encoding.**

Proposition 9.2 (Cost of the prime encoding)**.** Let $N(z) = \prod_i p_i^{d_i}$ with p_i the i-th prime and $d_i \in \{1, \dots, D\}$. Then:

(i) *Space.* $\log_2 N(z) = \sum_i d_i \log_2 p_i$. In the worst case — m curves active at maximal direction — this is $\Theta(D \cdot m \log m)$ bits, because the primorial satisfies

$$\sum_{i \le m} \log p_i \;=\; \vartheta(p_m) \;\sim\; p_m \;\sim\; m \ln m$$

by Chebyshev's function and the prime number theorem. Even at minimal direction the primorial factor $m \log m$ survives.

(ii) *Decode.* Curve i recovers d_i as the p_i-adic valuation $v_{p_i}(N)$: at most D divisions of the $\Theta(m \log m)$-bit value by the *known* prime p_i. This is *not* general integer factorisation — the primes are fixed and assigned to curves — so it is polynomial, $O\big(D \cdot \mathsf{M}(m \log m)\big)$ with M the big-integer multiplication cost, and not the sub-exponential cost of general factoring (the number field sieve is sub-exponential, but is never invoked here because the factor primes are known). Enumerating all m curves costs $O(m)$ valuations.

Remark 9.3 (The factoring worry is misplaced; the space cost is not)**.** The reader never factors a general integer — it divides by *its own* known prime — so the encoding is not "as hard as factoring". Its true costs lie elsewhere: the primorial-sized value forces $\Theta(m \log m)$ space, which is super-linear in the number of curves, and it forces big-integer arithmetic at every read.

9.3. **More compact separating families.** Against these costs stands an information-theoretic floor that any encoding must clear, and three realisations that clear it in different ways.

Proposition 9.4 (Information lower bound)**.** Any injective encoding of m independent symbols drawn from an alphabet of size $D + 1$ needs at least $m \log_2(D + 1)$ bits (Shannon source coding; see Cover–Thomas).

Proposition 9.5 (Mixed-radix / positional code — space-optimal)**.** Store $N = \sum_i d_i \, (D + 1)^{i-1}$ and decode $d_i = \big\lfloor N/(D+1)^{i-1} \big\rfloor \bmod (D + 1)$. The space is $m \log_2(D + 1)$ bits, meeting the bound of Proposition 9.4; all m curves decode by successive division in $O(m)$; per-curve access needs the weight $(D + 1)^{i-1}$. This is the positional / mixed-radix code, the Cantor tuple function (Cantor pairing; mixed-radix, Knuth, TAOCP vol. 2), and it admits unbounded m.

Proposition 9.6 (Residue number system / CRT — $O(1)$ per-curve access)**.** Assign curve i a modulus $q_i > D$, the q_i pairwise coprime, and store the $N < \prod_i q_i$ with $N \equiv d_i \pmod{q_i}$ by the Chinese Remainder Theorem. Decode $d_i = N \bmod q_i$ — a *single* modular reduction, $O(1)$ per curve, with no cumulative weights and no valuation. This is Garner's residue number system [Gar59]: the residues are independent channels, exactly the per-curve independence

(A4) that the problem asks for. Crucially, taking $q_i = p_i$ gives $N \equiv d_i \pmod{p_i}$ — the *same* primes as the paper's encoding, but used as *moduli* rather than as prime-power *bases*: the decode is one modular reduction instead of a valuation, and the value $N < \prod_i p_i$ is smaller than the prime-power $\prod_i p_i^{d_i}$ by up to a factor D in the exponent. Hence the prime-power encoding is *dominated* by CRT on both decode cost and value size, at equal generic properties (unbounded m, per-curve random access).

Proposition 9.7 (Bit-packing — the bounded-m practical optimum)**.** If $m \le K$ for a per-cell cap K (relaxing (A3)), pack K fields of $w = \lceil \log_2(D+1) \rceil$ bits: store $N = \sum_i d_i \, 2^{(i-1)w}$ and decode $d_i = (N \gg (i-1)w)$ AND $(2^w - 1)$. The space is Kw bits — one machine word when $Kw \le 64$ (e.g. $\delta = 2$, $D = 8$, $w = 4$: up to 16 curves per 64-bit word); the decode is shift-and-mask, $O(1)$ per curve on *small*, word-sized integers, with no big-integer arithmetic — the fastest of all.

When, in addition, the set of occurring curve-tuples is known and sparse, an FKS minimal perfect hash [FKS84] gives $O(1)$ access at near-optimal space — the bounded-known-keyset regime.

Table 2 places the four realisations side by side.

Encoding	**Space (bits)**	**Per-curve decode**	**All-curve decode**	**Unbdd. m?**	**$O(1)$ access?**	**Big-int?**	**δ-dim scaling**
Prime power $\prod_i p_i^{d_i}$	$\Theta(D\, m \log m)$	valuation	$O(m)$	yes	$\sim$	yes	blows up by D
RNS – CRT	$\Theta(m \log m)$	one mod, $O(1)$	$O(m)$	yes	**yes**	yes	linear in δ (via $\log D$)
Mixed-radix	$m \log(D{+}1)$ **(opt.)**	division	$O(m)$	yes	via weight	yes	linear
Bit-packing	$K\lceil \log(D{+}1) \rceil$, one word	shift+mask, $O(1)$	$O(K)$	no ($m \le K$)	**yes**	**no**	linear
FKS perfect hash	near-optimal	hash, $O(1)$	$O(m)$	known keyset	**yes**	no	linear

Table 2: Separating families for m superimposed curves, compared. All five realise the *same* separating family (Theorem 8.4) and so resolve every crossing; they differ only in the coordinate system on the direct sum $V \cong \prod_i D_i$. The prime-power form (Proposition 9.2) is *dominated*: RNS matches it on generic properties while decoding in $O(1)$ per curve and storing a smaller value. Mixed-radix (Proposition 9.5) is space-optimal, meeting the bound of Proposition 9.4; RNS (Proposition 9.6) gives $O(1)$ per-curve random access with unbounded m; bit-packing (Proposition 9.7) is the bounded-m practical optimum, one word and shift-and-mask.

9.4. **Arbitrary dimension, and the recommendation.**

Remark 9.8 (Arbitrary dimension)**.** On a δ-dimensional grid the Moore neighbourhood has $D = 3^\delta - 1$ directions ($\delta = 2$: 8; $\delta = 3$: 26; $\delta = 4$: 80), so $\log D = \Theta(\delta)$. Every separating family above extends by enlarging the per-curve alphabet to D; the space of the positional, RNS, and bit-packed forms then grows as $\Theta(m \cdot \delta)$ — linear in the dimension — whereas

the prime-power form carries the extra multiplicative $D = 3^\delta - 1$ blow-up in the exponent, a further reason it is not the efficient choice at higher dimension.

The prime-power encoding is therefore kept only as the unique-factorisation *illustration* of the separating family — its separation is visually obvious — but it is dominated in practice. Use *mixed-radix* for minimal space (it meets the information bound of Proposition 9.4); use the *residue number system* (CRT) for $O(1)$ per-curve random access with an unbounded curve count; and use *bit-packing* when the per-cell curve count is bounded (one word, shift-and-mask). All preserve the separating-family property and hence the resolution theorem (Theorem 8.4): efficiency is purely the choice of coordinate system on the same direct sum.

10. Local Encoding of Intersections: Routing and the Conditions of Possibility

The disentangling codes of Section 7 let several curves share a cell and be recovered by an observer who reads the whole cell value. We now ask the sharper, operational question that is the technical heart of this note: can a *tracer* — an agent walking one curve, one step at a time, using only the datum stored at the cell it currently occupies — always find its own successor at a shared cell? The answer is a clean dichotomy. The positive half is a genuine possibility theorem: with the right, and small, notion of *local context*, one integer per cell suffices to trace an entire family correctly. The negative half is its exact boundary. Along the way we settle a recurring temptation — that complex, Gaussian, or hypercomplex cell values buy tracing power — and show that they do not: they buy geometry, which is real, but not disambiguation, which is where the limit lies.

10.1. **Problem statement.** Fix one curve and imagine a tracer walking it. At each step the tracer sits on a cell, reads what is stored there, and must decide the next step from that purely local datum. When the cell is private to the curve this is the successor rule of Definition 6.2. The interesting case is a shared cell, where the tracer must still recover *its own* continuation. The constraints are:

(i) each cell stores a *single* number $N(z)$;
(ii) no explicit list $[(\text{curve_id}, \text{direction}), \dots]$ is stored;
(iii) several curves may share a cell;
(iv) even at an intersection the tracer must locally recover the successor of the curve it is following.

10.2. **Grid, directions, and curves over $\mathbb{Z}[i]$.** It is convenient here to name cells by Gaussian integers, because the eight Moore steps then become genuine ring elements and rotation becomes multiplication.

Definition 10.1 (Gaussian grid)**.** The *cells* are the Gaussian integers $z = x + iy \in \mathbb{Z}[i]$, with $x = \operatorname{Re} z$ and $y = \operatorname{Im} z$ the grid coordinates of Section 3.

Definition 10.2 (Direction set)**.** The *direction set* is

$$\mathcal{D} \;=\; \{\, \pm 1,\ \pm i,\ 1+i,\ 1-i,\ -1+i,\ -1-i \,\} \;\subset\; \mathbb{Z}[i],$$

the eight Moore steps. Multiplication by i is a 90° rotation, so $\mathcal{D}$ is closed under the rotation group $\{1, i, -1, -i\}$ — a genuine convenience of $\mathbb{Z}[i]$. A step is $z_{t+1} = z_t + \Delta z$ with $\Delta z \in \mathcal{D}$.

Definition 10.3 (Discrete curve)**.** A *curve* is a sequence (z_t) with $z_{t+1} - z_t \in \mathcal{D}$ for every t.

Definition 10.4 (Shared cell, intersection)**.** A cell z is *shared* by curves C and C' if z occurs in both. An *intersection* is a shared cell at which the two curves have different local geometry.

10.3. **Storage versus disambiguation.** Before proposing any encoding it pays to separate two problems that the phrase "one number per cell" silently conflates.

Remark 10.5 (The conceptual pivot)**.** Two distinct problems hide inside "one number per cell".

(P-store) **Storage.:** Fit k local instructions into one number.
(P-disamb) **Disambiguation.:** Given the tracer's local context, select the *one* instruction that is its own.

The claim we make precise below is that storage is solved by any injective pairing and is insensitive to the number system, whereas disambiguation is not a storage problem at all and is where the real limit lies.

10.4. **What complex numbers buy.** The geometry of the grid sits naturally in $\mathbb{Z}[i]$, and it is worth being exact about what that gains and what it does not.

Proposition 10.6 (Geometry is free in $\mathbb{Z}[i]$)**.** Position, the eight directions, stepping, and 90° rotation are all native operations of $\mathbb{Z}[i]$; encoding a single direction as an element of $\mathcal{D}$ costs $O(1)$.

Proof. A cell is an element $z \in \mathbb{Z}[i]$; a direction is an element $\Delta z \in \mathcal{D} \subset \mathbb{Z}[i]$; a step is the ring addition $z + \Delta z$; and a 90° turn is the ring multiplication $\Delta z \mapsto i\,\Delta z$, which permutes $\mathcal{D}$. Each is one constant-cost ring operation. □

Proposition 10.7 (No storage advantage from complex or hypercomplex values)**.** A Gaussian integer carries two integer channels $(\mathrm{Re}, \mathrm{Im})$, a quaternion four; but for storing a finite record of $(\text{curve}, \text{direction})$ pairs, any of these is inter-encodable with a single natural number by a fixed pairing (Cantor or Gödel), so no complex or hypercomplex algebra stores strictly more than $\mathbb{Z}$ at fixed cost. Reaching for quaternions to gain storage is therefore unjustified: routing eight-in/eight-out needs at most a table of 9^8 entries, i.e. about 25 bits — one integer.

Proof. A finite record of pairs is a finite sequence of naturals, which a fixed injective pairing (Cantor's, or Gödel's prime-power code of Definition 7.1) sends bijectively onto $\mathbb{N}$; the two channels of a Gaussian integer or the four of a quaternion are themselves finite tuples of integers and so likewise fold into a single natural. Hence every such algebra has exactly the storage capacity of $\mathbb{Z}$ at fixed cost. The eight-in/eight-out routing table has 9^8 entries, and $\log_2 9^8 = 8 \log_2 9 \approx 25.4$ bits. □

10.5. **The local tracer and the routing function.** The right object is not a richer cell value but an explicit model of the walking agent.

Definition 10.8 (Local deterministic tracer)**.** A *local deterministic tracer* is a tuple $(S, s_0, \mathrm{Val}, \mathrm{Decode})$ with S a finite state set, $s_0 \in S$ an initial state, Val the cell-value alphabet ($N(z) \in \mathrm{Val}$), and a total transition

$$\mathrm{Decode}\colon\ \mathrm{Val} \times S \times \mathcal{D} \ \longrightarrow\ (\mathcal{D} \cup \{\mathsf{stop}\}) \times S.$$

Operationally, a tracer at cell z_t in state s_t, having *arrived* via direction δ_t (so $z_t = z_{t-1} + \delta_t$), computes $(\delta_{t+1}, s_{t+1}) = \mathrm{Decode}(N(z_t), s_t, \delta_t)$ and steps $z_{t+1} = z_t + \delta_{t+1}$; it halts on stop.

Definition 10.9 (Cell as router)**.** Equivalently, $N(z)$ is the *index* of a partial routing function $f_z\colon S \times \mathcal{D} \to (\mathcal{D} \cup \{\mathsf{stop}\}) \times S$: the cell is a Mealy-machine transition table and its number is the Gödel index of that table. "One number per cell" costs nothing extravagant: a cell need not satisfy $|\mathrm{Val}| \geq |S| \cdot |\mathcal{D}|$; a single index into the $9^{|S|\cdot 8}$ possible tables suffices, one integer.

Remark 10.10 (The successor code is the $|S| = 1$ special case)**.** The self-descriptive Freeman curve of Section 6 is the stateless, entry-independent tracer: f_z ignores S and δ and returns the stored direction. Entry-direction routing — Decode depending on δ — is the first refinement; finite memory ($|S| > 1$) the second.

10.6. **Possibility: input-separated families are traceable.** This subsection carries the section's principal positive result: a precise, sufficient (and, by Section 10.7, necessary) condition under which a single integer per cell traces an entire family.

Definition 10.11 (Local signature of a passage)**.** When a tracer follows curve C through a shared cell z it arrives in some state s and via some entry direction δ; the pair (s, δ) is the *local signature* of that passage, and the required output is C's exit direction ε at z.

Definition 10.12 (Input-separated family)**.** A family of curves is *input-separated* (relative to a tracer) if the partial map $(z, s, \delta) \mapsto \varepsilon$ induced by all its passages is single-valued — a partial function.

Proposition 10.13 (Possibility)**.** If a family is input-separated then there is a single-number local encoding N and a deterministic tracer that traces every curve of the family correctly. If moreover separation is achieved by entry direction alone ($|S| = 1$), the tracer is stateless and $N(z) \in \{0, \ldots, 9^8 - 1\}$ encodes $f_z\colon \mathcal{D} \to \mathcal{D} \cup \{\mathsf{stop}\}$.

Proof. By Definition 10.12 the induced map $(z, s, \delta) \mapsto \varepsilon$ is a partial function. For each cell z set $f_z(s, \delta) = \varepsilon$ on the finitely many signatures that actually occur at z, and assign an arbitrary value (say stop) elsewhere; this is a well-defined total table because the map is single-valued. Let $N(z)$ be the index of f_z among the $9^{|S|\cdot 8}$ tables, and let Decode evaluate the indexed table. The tracer then reproduces each passage's exit direction by construction. When separation uses entry direction alone the state is inert, $|S| = 1$, and $f_z\colon \mathcal{D} \to \mathcal{D} \cup \{\mathsf{stop}\}$ is indexed by a single element of $\{0, \ldots, 9^8 - 1\}$. □

10.7. **The boundary: an impossibility theorem.** The possibility of Section 10.6 is sharp. Exactly one situation defeats every local tracer, and naming it converts the sufficient condition above into a characterisation.

Theorem 10.14 (Local indistinguishability)**.** *Let z be a cell shared by curves C and C'. Suppose that, at z, the two passages have the* same *tracer state and the* same *entry direction, $(s, \delta) = (s', \delta')$, but* distinct *exit directions $\varepsilon \neq \varepsilon'$. Then no local deterministic tracer $(S, s_0, \mathrm{Val}, \mathrm{Decode})$ — with cell-value alphabet* Val *of any cardinality and state set S of any finite size — traces both C and C' correctly.*

Proof. Decode is a total function, so $\mathrm{Decode}(N(z), s, \delta)$ has a unique value (ε'', s''). Tracing C correctly forces $\varepsilon'' = \varepsilon$; tracing C' correctly forces $\varepsilon'' = \varepsilon'$. Hence $\varepsilon = \varepsilon'$, contradicting $\varepsilon \neq \varepsilon'$. The value $N(z)$ is universally quantified and plays no role in the argument: no choice of cell number, in any number system, removes the contradiction. □

Corollary 10.15 (Characterisation)**.** *A family of curves is traceable by a single local deterministic tracer if and only if it is input-separated.*

Proof. Proposition 10.13 gives the "if", and Theorem 10.14 the "only if": a family that is not input-separated has, by Definition 10.12, a cell z with two passages of equal signature (s, δ) and distinct exits, which is exactly the hypothesis of Theorem 10.14. □

Corollary 10.16 (Where the resource must go)**.** *To trace a family that is not input-separated one must enlarge not the cell number but the* context*: (a) more tracer memory (larger S); (b) curve identity carried in S; (c) a phase/parity state; (d) non-determinism; (e) a global rule; or (f) geometric separation of the curves. Complex, Gaussian, and hypercomplex cell values are storage devices and are, by Theorem 10.14, powerless against a state-blind tangency.*

11. Worked Example and Related Systems: Robots on Grids

The dichotomy of Section 10 — one static integer per cell traces a curve family exactly when the family is input-separated (Corollary 10.15), and a state-blind tangency is resolved by no cell number (Theorem 10.14) — is easiest to feel on an explicit small grid. This closing subsection does two things. First (Section 11.1) it walks a *progressive* worked example that escalates the object to be represented — point, line, curve, crossing, plateau — until the escalation lands exactly on the impossibility theorem, and then names the two escapes. Second (Section 11.2) it positions the resulting static/mutable picture against the "robots on grids / one number per cell" literature, whose affirmative answer to the motivating question — *is there a numeric system for moving on a 2D grid with one number per cell?* — is the mutable rotor-router.

11.1. **A progressive worked example.** Each step raises the complexity of the object and asks what the single cell number must therefore carry. The spine is monotone: as the object grows, so does the resource the cell datum must supply — until, at a shared plateau, it cannot.

Example 11.1 (Point, segment, curve — static, stateless successor)**.** The three lowest cases share one verdict and are elementary. A *point* (an isolated occupied cell, Definition 4.1) marks presence with no successor; a *segment* or *line* carries a *constant* Freeman code

(Definition 6.1), every cell saying "keep going the same way". Let the object now bend: a curve stores a *varying* Freeman code — here $1, 1, 2, 3, 3$ (east, east, north-east, north, north) — so the successor depends on position. The requirement is now a *spatially-varying successor*, but the cell datum is still one static number and the tracer is still stateless (Remark 10.10). Figure 6 draws the bent path with its per-cell direction digits.

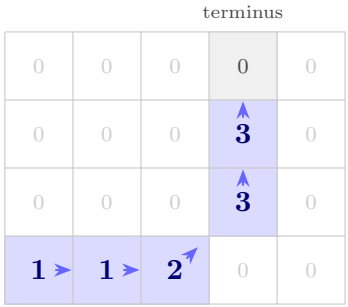


Figure 6: **E3 — curve.** A bent path with varying Freeman codes $1, 1, 2, 3, 3$ (east, east, north-east, north, north) reaching a terminus ($f = 0$); the successor now depends on position. Still one static number per cell, still a stateless tracer (Example 11.1).

Example 11.2 (Transverse crossing — entry-direction routing)**.** Now two curves cross at one cell with *different* entry directions: an east-goer arrives from the west and must leave east; a north-goer arrives from the south and must leave north. The cell encodes a routing function $f_z(\delta_{\text{in}}) = \delta_{\text{out}}$ with $f_z(\text{E}) = \text{E}$ and $f_z(\text{N}) = \text{N}$. The requirement is *entry-direction routing*, and because the two passages differ in entry direction the family is input-separated: one static number (the table index) and a stateless tracer resolve it, exactly the possibility case of Proposition 10.13. Figure 7 shows the crossing cell with its two in/out pairs.

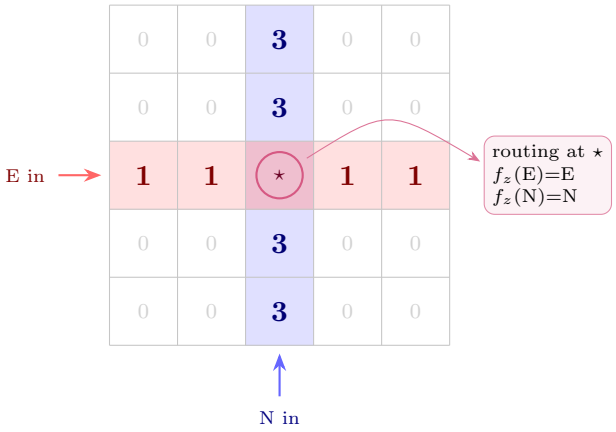


Figure 7: **E4 — transverse crossing.** Two curves cross at the purple cell with *different* entry directions: the east-goer (red, code 1) routes $\text{E} \mapsto \text{E}$, the north-goer (blue, code 3) routes $\text{N} \mapsto \text{N}$. The passages are input-separated by entry direction, so one static table index resolves the cell (Proposition 10.13 and Example 11.2).

Example 11.3 (Common plateau — the wall)**.** Escalate once more. Two curves *share* a run of cells — a plateau — entering its first cell from the *same* direction, traversing the run *identically*, and *diverging* only at the run's last cell: one exits east, the other north. At that branch cell the two passages present the *same* entry direction and, to a stateless tracer, the *same* state, yet demand *distinct* exits. This is exactly the hypothesis of the local-indistinguishability theorem (Theorem 10.14): *no* static single number, in any number system, disambiguates them. The escalation has hit the paper's impossibility wall. Figure 8 circles the branch cell — same entry, two exits.

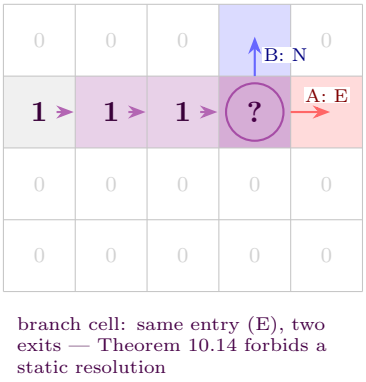


Figure 8: **E5 — common plateau (the wall).** Two curves enter the plateau (violet) from the west, traverse it identically (shared code 1, east), and diverge only at the circled branch cell: curve A exits east, curve B exits north. The two passages share entry direction and stateless state but demand distinct exits — the hypothesis of Theorem 10.14 — so no static single number disambiguates them.

Two escapes cross the wall, and only two: add state to the *tracer*, or add mutability to the *cell*.

Remark 11.4 (Escape 1 — resource in the context)**.** Give the tracer a way to tell the curves apart: a curve-identity tag or a single parity bit carried in its state s, so that the branch routes on $f_z(s, \delta)$ rather than on δ alone. One static cell number plus one bit of tracer state resolves the plateau — the "resource in the context" reading of Corollary 10.16. The cell datum stays static; the disambiguating resource lives in the walker.

Definition 11.5 (Escape 2 — rotor cell)**.** A *rotor cell* at z carries a *mutable* value $r(z) \in \mathbb{Z}/k\mathbb{Z}$ together with a fixed cyclic list of exits $(\varepsilon_0, \dots, \varepsilon_{k-1}) \in \mathcal{D}^k$. On each visit the cell outputs the direction $\varepsilon_{r(z)}$ and then advances $r(z) \leftarrow r(z) + 1 \bmod k$. Unlike the static index of Definition 10.9, the datum is *read-and-advance*: successive visits read successive rotor positions. This is the rotor of the rotor-router model [HLM$^+$08].

Proposition 11.6 (Plateau resolution by a rotor)**.** Let a plateau be shared by m curves that reach a common branch cell z with identical local signature (s, δ) but pairwise distinct required exits $\varepsilon_1, \dots, \varepsilon_m$. If the m curves cross z on distinct visits, then a rotor cell (Definition 11.5) of period $k = m$ at z, whose cyclic list is the exits in visit order, routes every curve to its own exit.

Proof. Number the visits $0, \ldots, m-1$ in order and let ε_j be the exit of the curve visiting at step j; the advance rule makes visit j read rotor position $j \bmod m$ and emit ε_j, so over one period each of the m exits is used exactly once (pigeonhole on $\mathbb{Z}/m\mathbb{Z}$). □

Remark 11.7 (Abelian well-definedness of the fan-out)**.** The rotor-router is *abelian*: the multiset of exit directions produced by all traversals of z is independent of the *order* in which the curves arrive [HLM$^+$08]. Hence the plateau's fan-out is well-defined even when the interleaving of the passages is unknown — whatever the arrival order, one full period distributes the m passages over the m exits, one each. The escape thus keeps "one number per cell," but the number is *mutable*.

The escalation makes the paper's dichotomy tangible. A *static* number per cell traces every input-separated family — points, lines, curves, and transverse crossings (Examples 11.1 and 11.2) — but hits the impossibility wall at a shared plateau (Example 11.3 and Theorem 10.14). The wall is escaped by adding state *either* to the tracer (context, Remark 11.4) *or* to the cell (a mutable rotor, Definition 11.5 and Proposition 11.6). The local-indistinguishability theorem is precisely the boundary of the static-cell model; the rotor-router is the mutable-cell model that crosses it, and crosses it *abelianly* (Remark 11.7).

11.2. **Related systems: robots on grids.** The routing/rotor picture sits inside a well-populated literature on *robots on grids* and *one number per cell*. Four families frame our contribution.

Rotor-router / Eulerian walkers / chip-firing. The closest system stores, at each vertex, a single mutable pointer — a rotor — that deterministically routes an incoming walker to the next outgoing edge in a fixed cyclic order; on a finite graph the walk settles into an Euler tour, and with many walkers or chips the outcome is *abelian*, independent of the interleaving of moves [HLM$^+$08]. Under its other names — Propp's "rotor-router machine" and the Eulerian-walkers model — this is exactly the mutable single-number-per-cell system that answers the research question in the affirmative. Our contribution is the complementary static-side result: the local-indistinguishability theorem (Theorem 10.14) that says *why* mutability, or equivalently tracer state, is necessary at a plateau.

Stigmergic ant-robots. A second family lets simple robots leave *evaporating* numeric traces in the cells; the mutable trace value guides a distributed swarm to provably complete coverage of an unknown grid, the decay preventing stale guidance [WLB99]. This is again a mutable-cell, multi-agent system — one number per cell, updated by passage — but its target is coverage rather than the disambiguation of crossing curves, so it sits on the same mutable-cell side of our boundary as the rotor-router.

Numerical navigation functions. A third family holds in each cell one *static* number equal to the distance-to-goal and descends it by steepest descent [BLL92]: the single-curve, gradient-flow static case, with no crossings or plateaus to disambiguate.

Space-filling / Morton codes, and path agents. A fourth family encodes cell *position* as one integer by bit-interleaving — the Morton or Z-order code *[Morton, IBM technical report, 1966 — cited textually, no reliable DOI]*, the "position as one number" leg complementary to our "direction as one number," while cellular-automaton path agents (Langton's ant / turmites *[Langton, Physica D 22, 1986 — cited textually]*) are our routing tracer read from the writing side.

Table 3 places the systems side by side. The pattern is uniform except at the two escapes of Theorem 10.14: the plain static-cell systems (navigation functions, Morton, Freeman) encode geometry but leave a plateau unresolved, while both escapes cross that boundary — the mutable rotor-router, abelianly for interleaved curves (Remark 11.7), and the static separating-family algebra with an indexed tracer (Theorem 8.4).

System	**Cell no.**	**Encodes**	**Curves**	**Disamb. crossings / plateaus?**	**Abel.?**	**Relation to this paper**
Our static routing cell	static	direction	multiple	only if input-separated (Theorem 10.14)	—	the boundary result
Rotor-router	mutable	rotor	multiple	yes — plateaus by rotor	yes	the mutable escape
Separating-family (CRT)	static	direct sum	multiple	yes — crossings & plateaus by projection	n/a	the static algebraic escape
Stigmergic ant-trace	mutable	trace	multiple (agents)	no — targets coverage	no	mutable-cell neighbour
Navigation function	static	distance	single (field)	n/a — one flow	n/a	single-curve static case
Morton / Z-order	static	position	n/a	no	n/a	the "position" leg
Freeman chain code	static	direction	single	no — one successor	n/a	the static base (Example 11.1)

Table 3: "Robots on grids / one number per cell" systems against the static/mutable boundary of this paper. **Cell no.**: is the single cell number static or mutable (read-and-advance). **Encodes**: what the number names. **Curves**: single flow or several interleaved curves. **Disamb.**: does the system resolve a shared crossing or plateau? **Abel.**: is the fan-out order-independent? Only the routing tracer resolves crossings on the static side (and only for input-separated families, Corollary 10.15); only the rotor-router resolves plateaus on the mutable side, abelianly (Remark 11.7); the separating-family cell resolves both crossings and plateaus statically once each tracer carries its curve index (Theorem 8.4), the static dual of the rotor. The disambiguation column follows from Corollary 10.15, Proposition 11.6, and Theorem 8.4; the remaining verdicts are design readings.

12. Conclusions

One static number per cell resolves m superimposed discrete curves, with each curve recovered locally at every shared cell and no collision, exactly when the cell's local-data

field factors through a separating family $V \cong \prod_i D_i$; the prime-per-curve code $N(z) = \prod_i p_i^{d_i(z)}$ realises it, each curve reading its direction off its own prime by unique factorisation (Theorem 8.4 and Definition 7.1). The result is sharp: the local-indistinguishability theorem (Theorem 10.14) shows that a state-blind tracer cannot resolve a tangency by any cell number, so the irreducible resource is the per-curve index, $\lceil \log_2 m \rceil$ bits, in the tracer.

The static/mutable duality. The impossibility theorem has exactly two escapes, and they are dual. The *separating family* keeps the cell a fixed algebraic superposition and puts the disambiguating resource in the tracer, as its channel index; the *rotor-router* keeps the tracer index-blind and puts the resource in the cell, as a mutable rotor advanced on each visit (Definition 11.5, Proposition 11.6, and Remark 11.7). Static cell with indexed tracer, or mutable cell with blind tracer: these are the only two ways across the wall.

A witness-region reading (internal connection). Within the program's region discipline (axis A1), the curve index is a *key*. An index-blind tracer holds no key and its passages sit in the key-dependent region R_4: whether it continues correctly through a tangency is not settled by any public reading of the cell. An index-carrying tracer holds the key and its reading is public and key-independent — region R_1: the projection $\pi_i(N(z))$ is a fact anyone with the index can check. The separating family is thus a witness-region statement about representation — the promotion $R_4 \to R_1$ is exactly the supply of the per-curve index.

Future work. The efficiency analysis (Section 9) already replaces the prime code with dominated-free realisations — space-optimal mixed-radix, $O(1)$-random-access residue (CRT), and one-word bit-packing — all sharing the separating-family property; three questions remain open beyond it. First, a *minimal-*$|V|$ question: when the data sets D_i and the overlap pattern of the curves are known, which encoding uses the fewest cell values — a coding-theory optimisation below the generic bound $|V| = \prod_i |D_i|$. Second, a *pencil* question: when the crossing structure is an algebraic pencil with base points of higher multiplicity, does the two-channel datum (f, g) still separate the members, or is a higher-rank family forced. Third, a *continuous-limit* question: as the grid refines, does the separating-family field converge to a section of a rank-m vector bundle whose fibres are the curves.

Self-intersecting curves: branch separation by residue jets. A self-intersection — a cell with two successors for one curve — is out of scope (Remark 6.5); the remedy is to index the residue channels by the local *branches* rather than by the curve: one further coprime modulus $q_i' > D$ per extra passage, so that $N \equiv d \pmod{q_i}$ and $N \equiv d' \pmod{q_i'}$ — the (in, out) datum of Definition 8.1 realised by residues. Algebraically the passages are the *branches* of the curve at the singular cell, and separating them is the *normalisation* of the local ring $\mathcal{O}$ into the product $\widetilde{\mathcal{O}} = \prod_j \mathcal{O}_j$ of its branch rings; the residue splitting $\mathbb{Z}/(q_i q_i') \cong \mathbb{Z}/q_i \times \mathbb{Z}/q_i'$ is the arithmetic mirror of that product. Two branches are separated by finitely much data exactly at the order of their mutual *contact*: at an ordinary node (transversal crossing) the tangents differ, so the branch is fixed by its tangent — the tracer's entry direction, the 1-jet, one extra residue. When the branches are *tangent* (a tacnode of contact order $\delta \geq 2$) the 1-jet coincides and one must read the δ-jet: residues modulo higher powers, the associated graded of $\mathcal{O}$ up to the delta-invariant $\dim \widetilde{\mathcal{O}}/\mathcal{O}$. The local-indistinguishability theorem (Theorem 10.14) is the limit of this tower — at a genuine tangency the branches agree to every order a finite-state tracer can read, no static residue jet separates them, and the resolving datum is the visit index carried in the tracer (Corollary 10.16). This poses an

open problem: characterise the minimal residue jet — how many channels, at which moduli, indexed by the branch jets up to the local contact order — that statically separates a given arrangement's branches, and bound the total channel count by the sum of delta-invariants (a conductor bound) and the arithmetic genus of the arrangement.

Intrinsic direction semantics (curvature coding). The datum d_i has so far been an *absolute* exit direction. Two refinements of its *semantics* — orthogonal to the separating-family algebra, which is untouched — deserve study. (i) An *angular* reading interprets the direction index as a phase, $\theta = 2\pi d_i/m$, and the step as $\mathrm{Q}(e^{\mathrm{i}\theta})$ with Q a quantiser onto the neighbourhood; at $m = 8$ this merely re-labels the Moore code, and for $m > 8$ the map Q onto an 8-neighbourhood is many-to-one, so the extra resolution is a silently excluded interval (DIA) unless sub-cell phase is accumulated — at which point the tracer leaves the finite-state regime, crossing `finite`→`bridge`. (ii) More promising, a *relative* reading stores a discrete *turning* $\Delta\theta_i = 2\pi d_i/m$ and lets the tracer carry its heading as a unit $\zeta_m^{\cdot} \in \mathbb{Z}[\zeta_m]$ (a cyclotomic integer, $\zeta_m = e^{2\pi\mathrm{i}/m}$; $\zeta_4 = \mathrm{i}$ recovers the Gaussian units), the cell acting as a local rotation $h \mapsto h\,\zeta_m^{d_i}$. This is the intrinsic, rigid-motion-invariant description of a curve by its discrete curvature (turning angle) — the differential (first-difference) refinement of the chain code [Fre61], the discrete analogue of the fundamental theorem of plane curves, and the turtle/L-system turning rule — and it is the natural home for a discrete-smoothness notion (bounded $|\Delta\theta|$). Two cautions: the turning index should ride the *efficient* channels of Section 9 (a bit-field or residue), not the dominated prime exponents, since algebra and direction-semantics are independent axes; and curvature coding does *not* escape Theorem 10.14 — a tangency still requires the per-curve index. The absolute-angular re-labelling is a convention with no new expressive power at $m = 8$; the relative/cyclotomic curvature reading is the substantive direction.

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